# ONBRD-10: Building Dashboards

> **Series:** ONBRD — Dynatrace Onboarding | **Notebook:** 10 of 10 | **Created:** December 2025 | **Last Updated:** 10/27/2026

## Visualizing Your Data

Dashboards provide at-a-glance visibility into your environment's health and performance. This notebook covers dashboard creation, common visualization patterns, and sharing with your team.

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## Table of Contents


1. Dashboards vs Notebooks
2. Creating Your First Dashboard
3. Tile and Visualization Types
4. Dashboard Patterns
5. Useful Queries for Dashboards
6. Sharing and Permissions
7. Next Steps


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## Prerequisites

- Viewer access or higher
- Familiarity with DQL (ONBRD-08)
- Data in your environment to visualize

# 1. Dashboards vs Notebooks

Dynatrace offers two visualization tools:

| Feature | Dashboard | Notebook |
|---|---|---|
| **Purpose** | Monitoring at a glance | Interactive analysis |
| **Layout** | Fixed grid tiles | Sequential document |
| **Auto-refresh** | Yes | Manual |
| **Interactivity** | Click-through links | Query editing |
| **Sharing** | Wall displays, reports | Investigation documentation |
| **Best for** | NOC screens, status pages | Troubleshooting, exploration |

This notebook uses **Dashboards**, the Latest Dynatrace app. Dashboards Classic is the older surface, and Dynatrace's advice for it is: *"If you're still using classic dashboards, we encourage you to upgrade your dashboards and benefit from all the latest dashboarding possibilities made available by the Dashboards app in the latest Dynatrace."* Classic tiles such as **Host health** and **Service health** do not exist in the Dashboards app; their job is done with DQL tiles (§5).

> **Sources:** Dashboards Classic (DT docs).

## When to Use Each

| Scenario | Use |
|---|---|
| Team status display | Dashboard |
| Incident investigation | Notebook |
| Executive summary | Dashboard |
| Root cause analysis | Notebook |
| SLA reporting | Dashboard |
| Ad-hoc queries | Notebook |

# 2. Creating Your First Dashboard

**Location:** the **Dashboards** app → create a dashboard

## Dashboard Creation Steps

1. Click "Create dashboard"
2. Name your dashboard (e.g., "Production Overview")
3. Add tiles by clicking "+" or dragging
4. Configure each tile
5. Arrange and resize as needed
6. Save

## Dashboard Settings

| Setting | Description |
|---|---|
| **Name** | Dashboard title |
| **Time frame** | Default time range |
| **Segment** | Default filter scope |
| **Variables** | *"Configure variable filters to monitor different resources within a single dashboard."* |
| **Refresh rate** | Auto-refresh interval. *"When you open a dashboard for the first time, the refresh rate is set to Off (no automatic refresh)."* A rate you pick is remembered the next time you open the dashboard |
| **Owner** | Dashboard owner |
| **Sharing** | Who can view/edit (§6) |

> **Sources:** Dashboards (DT docs).

# 3. Tile and Visualization Types

## Tiles

| Tile Type | Use Case |
|---|---|
| **Query** | A DQL query against Grail, shown with any visualization below |
| **Explore** | Point-and-click data exploration, no DQL needed |
| **Code** | Data returned by code run as a Dynatrace function |
| **Markdown** | Documentation, links |
| **Image** | Logos, diagrams |
| **Service-Level Objective** | SLO status |
| **Variables** | Filters that drive the other tiles |

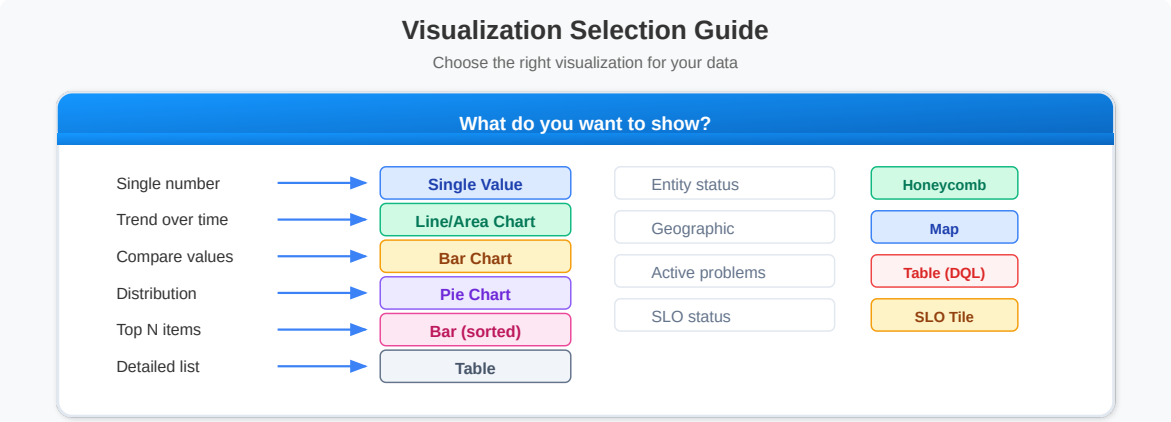
## Visualizations for Query and Explore Tiles

| Visualization | Use Case |
|---|---|
| **Single value** | Key metrics (request count, error rate) |
| **Line / Area / Band chart** | Time series data |
| **Bar / Categorical chart** | Comparisons and ranked top-N (sort and `limit` in the query) |
| **Table / Record list** | Lists and details, including active problems from `dt.davis.problems` |
| **Pie / Donut** | Distribution |
| **Honeycomb** | Entity health grid |
| **Meter bar / Gauge** | Value against a range |
| **Maps** (choropleth, dot, connection, bubble) | Geographic data |

There is no "top list" visualization: a ranked list is a bar or categorical chart (or a table) over a query that ends in `sort … | limit N`.

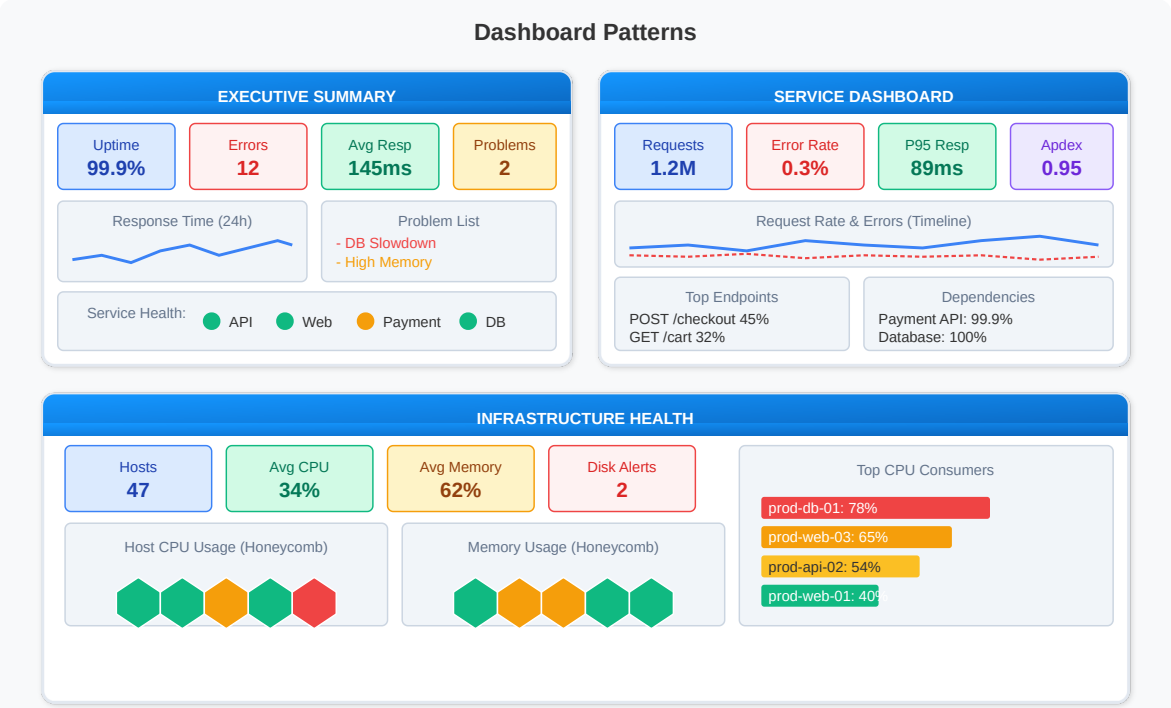
> **SaaS 1.349 (staged tenant rollout from 10/27/2026):** *"Top list visualizations are now integrated into Dashboards and Notebooks"* (What's new in Dynatrace SaaS 1.349 (DT docs)). Verify it has reached your tenant; until then the current guidance remains the working path.

## Visualization Types




> **Sources:** Dashboards (DT docs) — tile types; Edit visualizations (DT docs) — the visualization list.

# 4. Dashboard Patterns




## Pattern: Executive Summary

Key elements:
- **KPI row**: Uptime, Errors, Avg Response, Active Problems
- **Trend chart**: Response time over 24h
- **Problem list**: Current issues
- **Service health**: Status indicators for key services

## Pattern: Service Dashboard

Key elements:
- **KPI row**: Requests, Error Rate, P95 Response, Apdex
- **Timeline**: Request rate and errors over time
- **Top endpoints**: Most called API endpoints
- **Dependencies**: Downstream service health

## Pattern: Infrastructure Dashboard

Key elements:
- **KPI row**: Host count, Avg CPU, Avg Memory, Disk Alerts

- **Honeycomb views**: Visual host health for CPU and memory
- **Top consumers**: Hosts using most resources

# 5. Useful Queries for Dashboards

These queries work well as dashboard tiles. Each sets its own timeframe with `from:`, and *"If the timeframe is defined in the query itself, the dropdown list is disabled."* Remove `from:` when the tile should follow the dashboard timeframe.

> **Sources:** Dashboards (DT docs).

## Service Health Queries

The service tiles read the service request metrics `dt.service.request.count` and `dt.service.request.failure_count` rather than counting spans. Counting `span.kind == "server"` spans with `span.status_code == "error"` gives a different and misleading number. A service request is its request root span, which is not always a server span, and *"a request counts as failed only when failure detection marks it as such."* On a validation tenant (1 hour, 10/02/2026) the span version read an error rate of 0.35 % (407 of 115,103 server spans); the service metrics read 4.61 % (4,807 failures in 104,191 requests).

> **Sources:** Service-related concepts (DT docs).

```
// Total request count (Single Value tile)
timeseries requests = sum(dt.service.request.count, scalar: true), from: now()
- 1h
```

```
// Error rate percentage (Single Value tile)
timeseries {
    requests = sum(dt.service.request.count, scalar: true),
    failures = sum(dt.service.request.failure_count, scalar: true)
  }, from: now() - 1h
| fieldsAdd error_rate = round(100.0 * failures / requests, decimals: 2)
```

```
// Top services by request count (Bar chart tile)
timeseries requests = sum(dt.service.request.count, scalar: true), by:
{dt.service.name}, from: now() - 1h
| sort requests desc
| limit 10
```

```
// Service error rates (Table tile)
timeseries {
    requests = sum(dt.service.request.count, scalar: true),
    failures = sum(dt.service.request.failure_count, scalar: true)
  }, by: {dt.service.name}, from: now() - 1h
| fieldsAdd error_rate = round(100.0 * failures / requests, decimals: 2)
| sort error_rate desc
| limit 15
```

## Infrastructure Queries

```
// Host count (Single Value tile)
fetch dt.entity.host
| filter state == "RUNNING"
| summarize host_count = count()

// Smartscape note (dt.entity.* is deprecated but still functional): this
query uses the
// classic-only field state, which has NO Smartscape node equivalent
// (Smartscape expresses liveness via node lifetime, not a state field). Keep
the classic
// query above for state detail. For monitoring mode, do not use
monitoringMode — it is
// empty on Kubernetes and Fargate hosts; read billing events instead (ONBRD-
05 § 6).
```

```
// Hosts by OS type (Pie Chart tile)
fetch dt.entity.host
| summarize count = count(), by: {osType}
| sort count desc

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "HOST"
//   | summarize count = count(), by: {os.type}
//   | sort count desc
// Caveat: Smartscape reflects CURRENT live topology and can report fewer
entities
// than the classic entity store; for a pre-migration discovery inventory keep
the
// classic query above.
// Field maps: osType -&gt; os.type (LINUX -&gt; OS_TYPE_LINUX).
```

```
// Host inventory (Table tile)
fetch dt.entity.host
| fields name = entity.name, os = osType, state, cpuCores
| sort name
| limit 25

// Smartscape note (dt.entity.* is deprecated but still functional): this
query uses the
// classic-only field state, which has NO Smartscape node equivalent
// (Smartscape expresses liveness via node lifetime, not a state field). Keep
the classic
// query above for state detail. For monitoring mode, do not use
monitoringMode — it is
// empty on Kubernetes and Fargate hosts; read billing events instead (ONBRD-
05 § 6).
// Other fields do map: osType -&gt; os.type (LINUX -&gt; OS_TYPE_LINUX);
cpuCores -&gt; cores; entity.name -&gt; name.
```

## Log Queries

Filter error logs on `status`, not `loglevel`. `loglevel` is the source's own severity, while *"for each log event, a status attribute is created with a value that is a sum of loglevel values"*, and the levels `SEVERE`, `ERROR`, `CRITICAL`, `ALERT`, `FATAL` and `EMERGENCY` all map to `status == "ERROR"`. Java applications log `SEVERE`, so `loglevel == "ERROR"` misses them. On a validation tenant (1 hour, 10/02/2026) `status == "ERROR"` returned 118,086 records and `loglevel == "ERROR"` 63,551; most of the difference was 54,535 `SEVERE` records.

> **Sources:** Automatic log enrichment (DT docs).

```
// Error log count (Single Value tile)
// status groups SEVERE, ERROR, CRITICAL, ALERT, FATAL and EMERGENCY; loglevel
is the raw source value
fetch logs, from: now() - 1h
| filter status == "ERROR"
| summarize error_count = count()
```

```
// Log volume by severity (Pie Chart tile)
fetch logs, from: now() - 1h
| summarize count = count(), by: {loglevel}
| sort count desc
```

```
// Recent errors (Table tile)
fetch logs, from: now() - 1h
| filter status == "ERROR"
| fields timestamp, loglevel, log.source, content
| sort timestamp desc
| limit 10
```

## Problem Queries

```
// Active problem count (Single Value tile)
fetch dt.davis.problems, from: now() - 30d
| filter event.status == "ACTIVE"
| summarize active_problems = count()
```

```
// Problems by status (Pie Chart tile)
fetch dt.davis.problems, from: now() - 7d
| summarize problem_count = count(), by: {event.status}
```

```
// Recent problem list (Table tile)
fetch dt.davis.problems, from: now() - 24h
| fields timestamp, display_id, event.name, event.status
| sort timestamp desc
| limit 10
```

# 6. Sharing and Permissions

## Sharing Options

| Sharing Level | Who Can Access |
|---|---|
| **Private** | Only you, the owner |
| **Specific users or groups** | Named users and groups, each with *Can view* or *Can edit* |
| **Access for all** | *"let everyone in your Dynatrace environment view the document"* (view-only) |
| **Share link** | Anyone in the environment who has the link, with the permission chosen when the link was created |

A share link can be forwarded: *"anyone in your Dynatrace environment could use it, and they would have the same permissions (Can edit or Can view) that you selected when you created the link."* Use it for view access, not edit.

## Segment Filtering

Dashboards can be filtered using segments:
- Set a default segment in dashboard settings
- Viewers can switch segments if they have access
- Data respects viewer's permissions

## One Dashboard, Several Views

The Dashboards documentation describes no saved presets. To give one dashboard several views, use **Variables** (viewer-selectable filters such as service, cluster or environment) and a **Default segment**. The documentation's rule of thumb: use segments for filters you reuse across dashboards, and variables *"if you need more control over how a filter is applied"*.

### Exporting Dashboards

| Format | Use Case |
|---|---|
| **JSON** (*Download*) | Backup, migration: *"Download writes the dashboard to a JSON file that you can import"* |

> **Edited or generated JSON can stop displaying (SaaS 1.346, staged rollout from 08/25/2026).** *"Starting with Dynatrace version 1.346, Dynatrace applies stricter validation rules to dashboards and won't display dashboards that fail validation until you fix them."* It *"will mainly affect dashboards created or modified via the API or external AI tools."* Import edited or generated JSON into a test environment first and confirm it displays (DASH-07). Tenants not yet on 1.346 still display such dashboards.
>
> **Sources:** Share documents (DT docs); Dashboards (DT docs) — segments, variables and Download; SaaS 1.346 (DT docs).

## 7. Next Steps

Congratulations! You've completed the onboarding series.

### What You've Learned

| Notebook | Key Topics |
|---|---|
| **ONBRD-01** | Tenant access, navigation |
| **ONBRD-02** | IAM, SAML, Platform Tokens, parameterized policies |
| **ONBRD-03** | ActiveGate deployment |
| **ONBRD-04** | Cloud & SaaS integrations |
| **ONBRD-05** | OneAgent deployment, primary tags at source |
| **ONBRD-06** | Tags, segments, naming conventions, `dt.security_context` |
| **ONBRD-07** | Data model, topology |

| Notebook | Key Topics |
|---|---|
| **ONBRD-08** | DQL fundamentals |
| **ONBRD-09** | Workflows alerting |
| **ONBRD-10** | Dashboard creation |

## Where to Go Next — by Topic

| Goal | Topic Series |
|---|---|
| **Dashboard strategy + executive reporting** | DASH series (8 notebooks) |
| **Davis AI / anomaly detection / RCA** | AIOPS series (8 notebooks) |
| **Workflow automation + AI tasks** | WFLOW series (12 notebooks) |
| **Alerting strategy and design** | ALERT series (5 notebooks) |
| **Service level objectives** | SLO series (6 notebooks) |
| **Deepen DQL — spans / logs / OpenPipeline** | SPANS, OPLOGS, OPMIG, OPIPE |
| **Synthetic monitoring** | SYNTH series |
| **Web RUM / Mobile RUM** | WEBRUM, MOBL |
| **Business events & funnel analysis** | BIZEV series |
| **Database monitoring** | DBMON series |
| **Cloud integration deep dives** | CLOUD series |
| **Kubernetes monitoring** | K8S series |
| **OpenTelemetry integration** | OTEL series |
| **IAM administration depth** | IAM series |
| **Configuration automation / GitOps** | AUTOM series |

| Goal | Topic Series |
|---|---|
| **Access control modernization** | MZ2POL series (Management Zones → Policies) |
| **Migration playbooks** | NRLC + NR2DT (New Relic), SL2DT (Sumo), S2D (Splunk), M2S (Managed → SaaS) |
| **Platform maturity / adoption** | ADOPT series |
| **Official training** | Dynatrace University |
| **Community** | community.dynatrace.com |

### Dashboard Checklist

- [ ] First dashboard created (using **Dashboards (new)**, not legacy Classic Dashboards)
- [ ] Key metrics visualized
- [ ] Problem overview included
- [ ] SLO / SLI tile evaluated for service-health views
- [ ] Dashboard shared with team
- [ ] Segment filter set

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## Summary

In this notebook, you learned:

- Difference between Dashboards (new) and Classic Dashboards
- Difference between dashboards and notebooks
- How to create and configure dashboards
- Tile and visualization types and when to use them
- Dashboard patterns for different use cases
- DQL queries that work well in dashboards, including why service tiles read the service metrics and error-log tiles filter on `status`
- That `dt.davis.problems` queries use `event.status` (`ACTIVE` / `CLOSED`), not `status` (`OPEN`)
- How to share dashboards with your team

## References

- Dashboards (DT docs)
- Dashboards and notebooks (DT docs)
- Edit visualizations (DT docs)
- Share documents (DT docs)
- Dashboards Classic (DT docs)
- Notebooks (DT docs)
- Dynatrace Query Language (DT docs)
- Dynatrace Community

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